Assignment 01

Course No: MAT110 **Semester: Fall 2025** **Total Marks: 15**
Last Day of Submission: 2nd November, 2025.

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1. Find the limit of the following functions: (1×5)
   a. $\lim_{x \to 3^-} \frac{1}{|x-3|}$
   b. $\lim_{x \to 5} \frac{x^2-3x-10}{x^2-10x+25}$
   c. $\lim_{x \to +\infty} \left(\sqrt{x^6+5x} - x^3\right)$
   d. $\lim_{y \to 6} \frac{y+6}{y^2-36}$
   e. $f(x) = \begin{cases} \frac{1}{x+2} & , \quad x < -2 \\ x^2 - 5 & , \quad -2 < x < 3 \\ \sqrt{x+13} & , \quad x > 3 \end{cases}$ at $x = 3$.

2. Let $f(x) = |x| + |x-1|$. Find the limit and test the continuity of the function $f(x)$ at $x = 0$ and $x = 1$. (3)

3. $f(x) = \begin{cases} \frac{sin^2 bx}{x^2} & , \quad \text{if } x \neq 0 \\ 2 & , \quad \text{if } x = 0 \end{cases}$ (3)
   Prove that if $b = 2$, then the function is continuous at $x = 0$.

4. Let $f(x) = \frac{x-1}{\sqrt{x^2}}$ and $\lim_{x \to +\infty} f(x) = a$ and $\lim_{x \to -\infty} f(x) = b$. (4)
   The straight lines $y = a$ and $y = b$ are called Horizontal Asymptotes of the graph of $y = f(x) = \frac{x-1}{\sqrt{x^2}}$.
   Find the value of $(a - b)$.